Finding the Roots of Non-Linear Equations

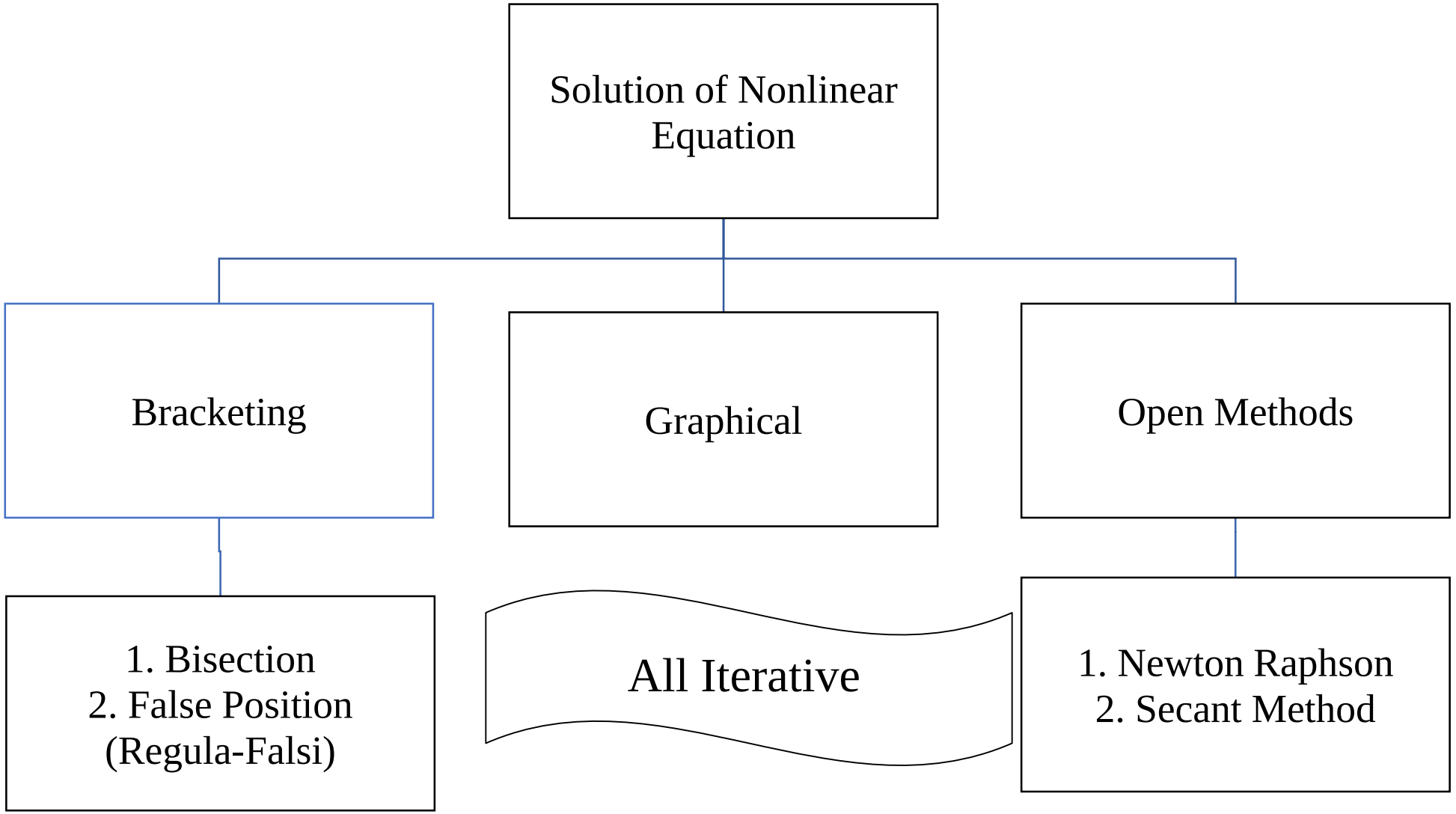
- Why? We find the root mathematically/analytically

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- But

$$ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0 \quad \Rightarrow x = ?$$

$$\sin x + x = 0 \quad \Rightarrow x = ?$$


Solution of Nonlinear Equation
Bracketing
Graphical
Open Methods
1. Bisection
2. False Position (Regula-Falsi)
All Iterative
1. Newton Raphson
2. Secant Method

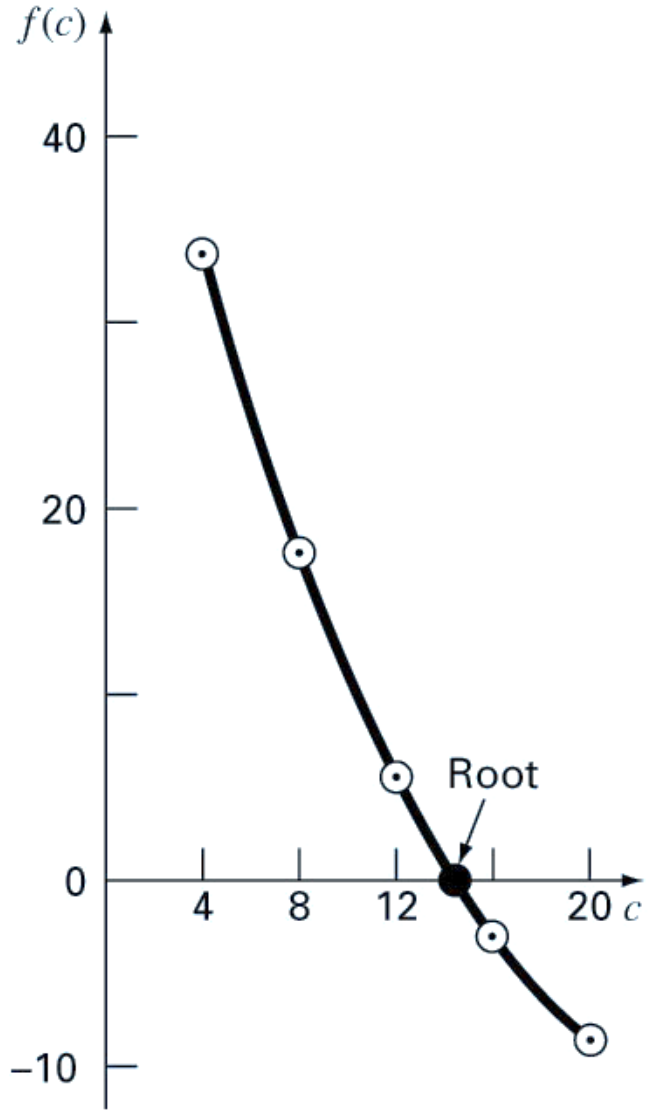
Bracketing Methods

(Or, two point methods for finding roots)

- Two initial guesses for the root are required. These guesses must "bracket" or be on either side of the root.

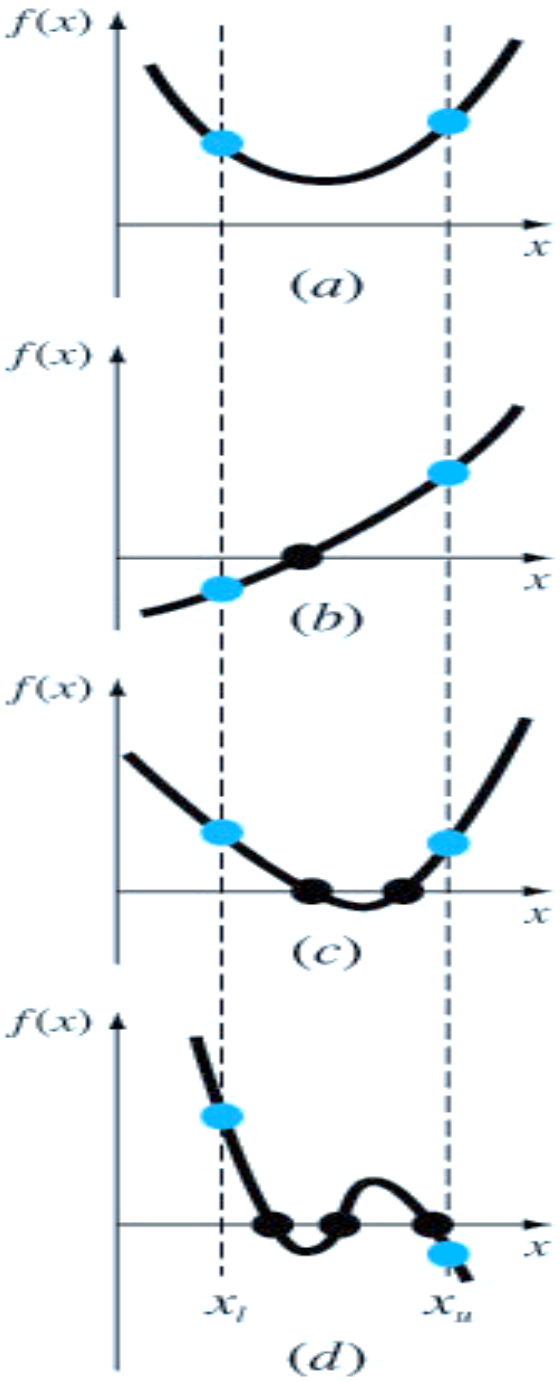
== > Fig

- If one root of a real and continuous function, $f(x)=0$, is bounded by values $x=x_l$, $x=x_u$
- Then $f(x_l) \cdot f(x_u) < 0$. (The function changes sign on opposite sides of the root)



Parts (a) and (c) indicate that if both $f(x_l)$ and $f(x_u)$ have the same sign, either there will be no roots or there will be an even number of roots within the interval.

Parts (b) and (d) indicate that if the function has different signs at the end points, there will be an odd number of roots in the interval.



No answer (No root)

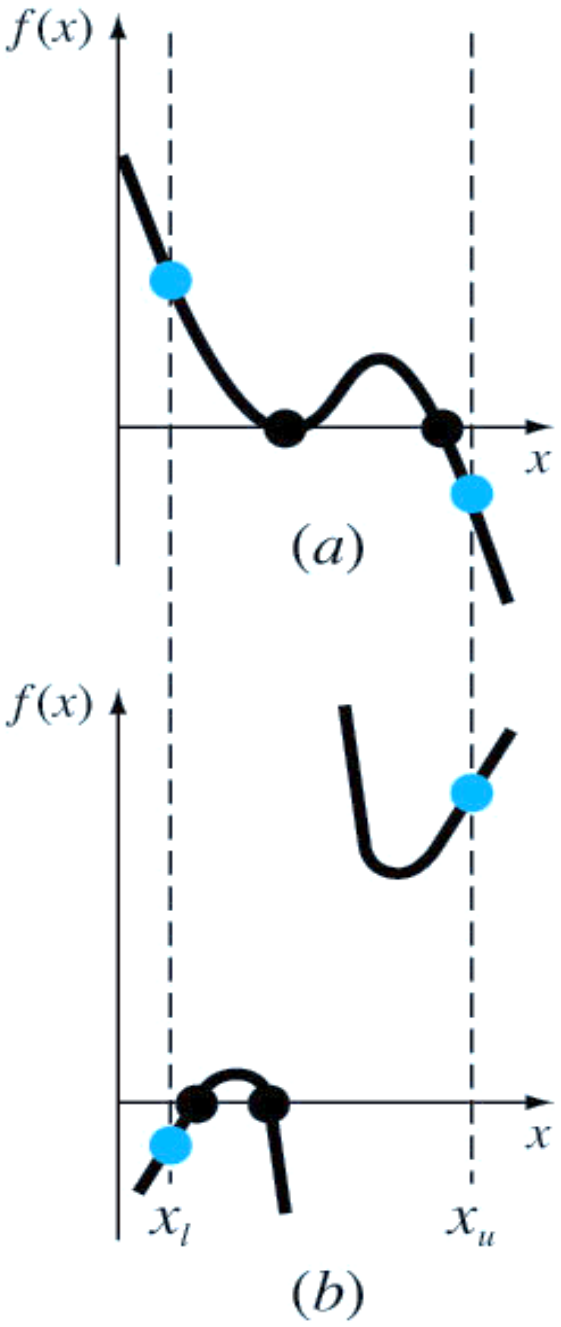
Nice case (one root)

Oops!! (two roots!!)

Three roots (Might work for a while!!)

(a) Multiple roots that occur when the function is tangential to the *x*-axis. For this case, although the end points are of opposite signs, there are an even number of axis interceptions for the interval.

(b) Discontinuous functions where end points of opposite sign bracket an even number of roots. Special strategies are required for determining the roots for these cases.



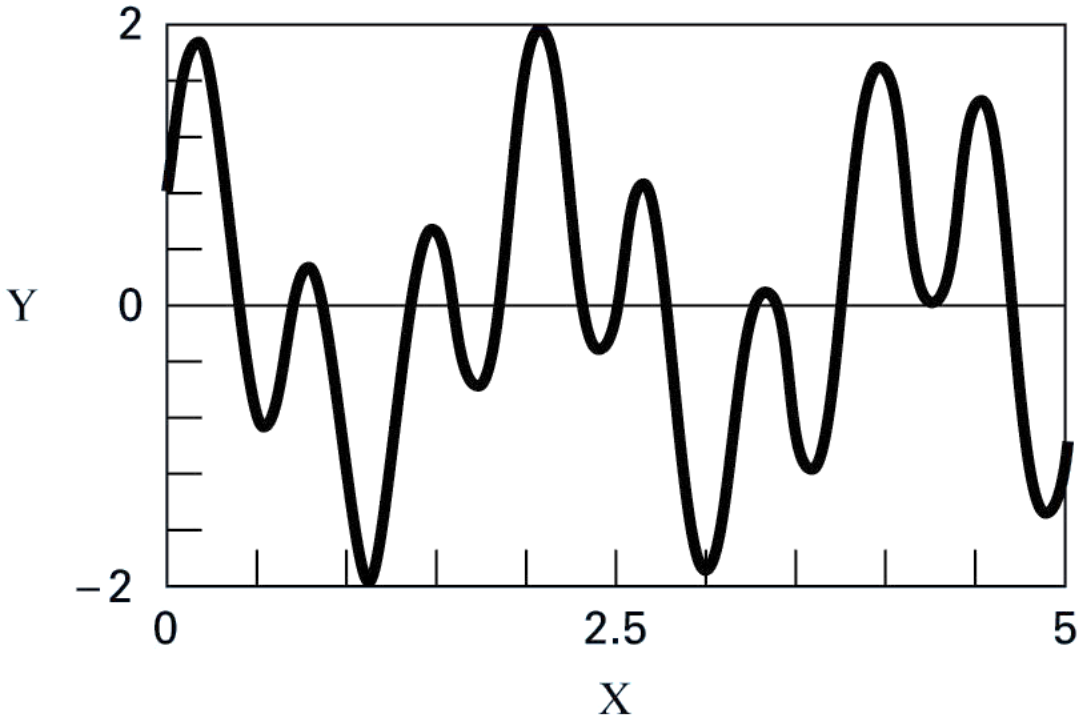
Two roots(Might work for a while!!)

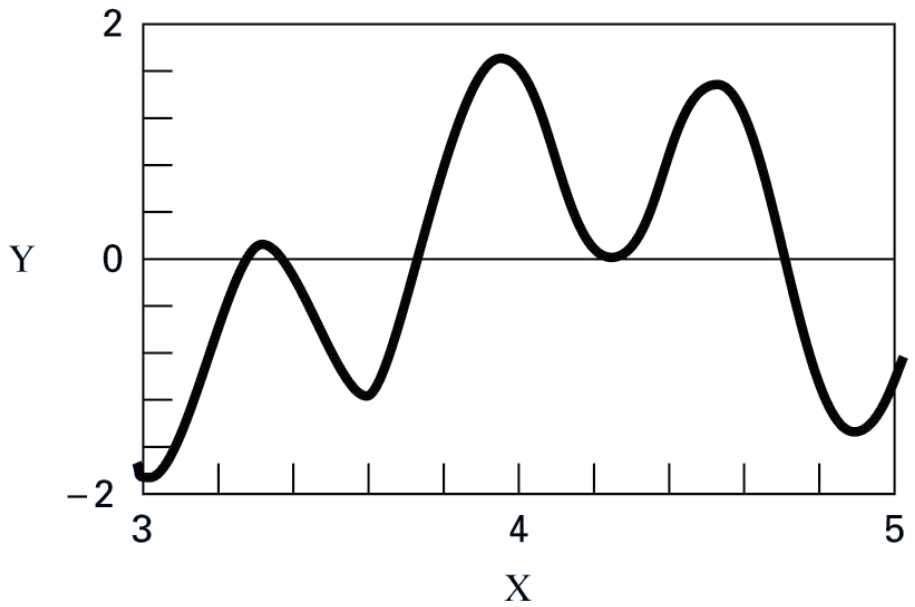
Discontinuous function. Need special method

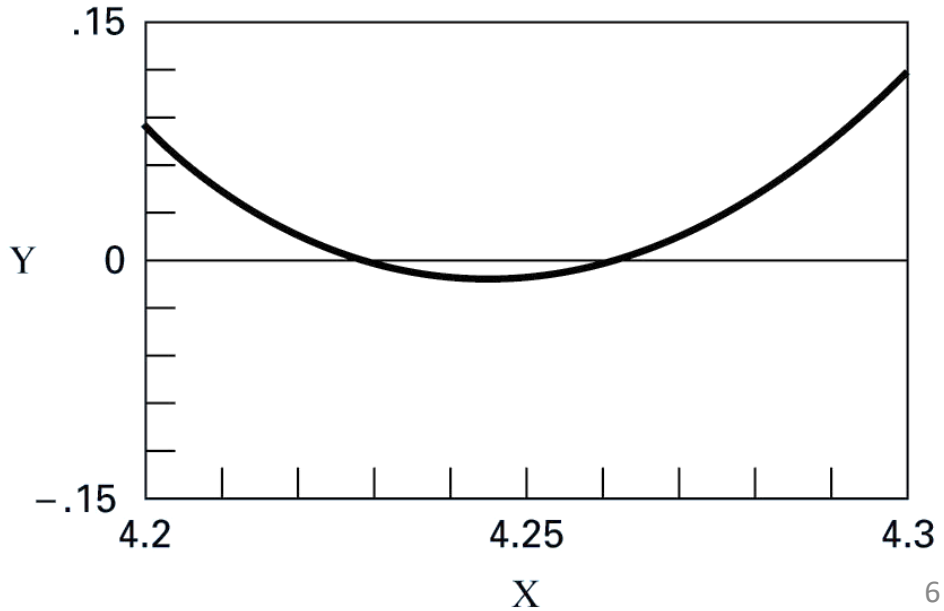
MANY-MANY roots.

What do we do? Incremental search

$$f(x) = \sin(10x) + \cos(3x)$$







The Bisection Method

For the arbitrary equation of one variable, $f(x) = 0$

1. Pick x_l and x_u such that they bound the root of interest, check if $f(x_l) \times f(x_u) < 0$.

2. Estimate the root by evaluating $f[(x_l+x_u)/2]$.

3. Find the pair

   - If $f(x_l) \times f[(x_l+x_u)/2] < 0$, root lies in the lower interval, then $x_u = (x_l+x_u)/2$ and go to step 2.

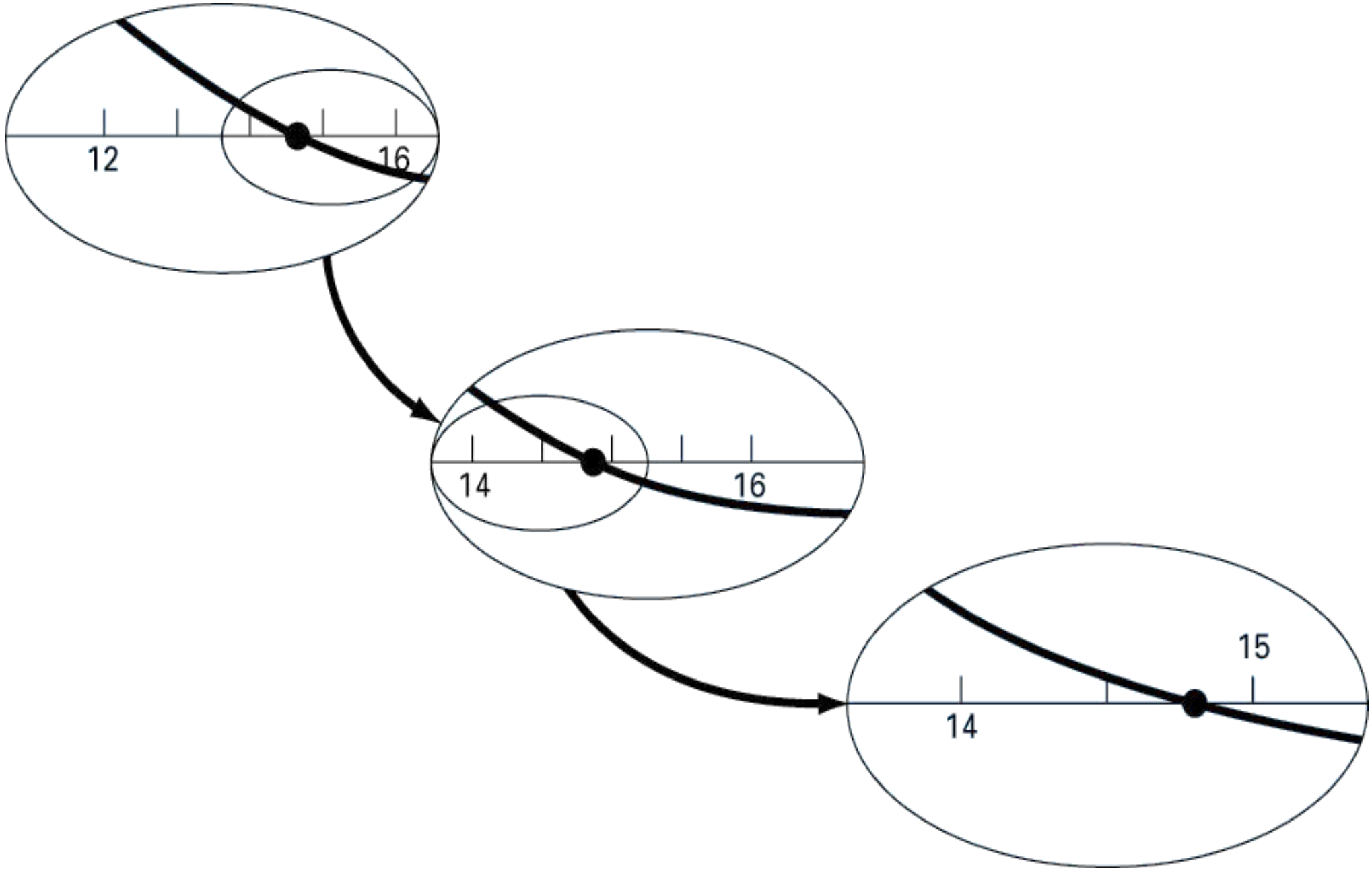
- If $f(x_l) \times f[(x_l+x_u)/2] > 0$, root lies in the upper interval, then $x_l = [(x_l+x_u)/2$, go to step 2.

- If $f(x_l) \times f[(x_l+x_u)/2]=0$, then root is $(x_l+x_u)/2$ and terminate.

- Compare ε_s with ε_a

- If $\varepsilon_a < \varepsilon_s$, stop. Otherwise repeat the process.

$$\left\{ \begin{array}{l} \dfrac{\left| x_l - \dfrac{x_l + x_u}{2} \right|}{\left| \dfrac{x_l + x_u}{2} \right|} \prec 100\% \\ \textit{or} \\ \dfrac{\left| x_u - \dfrac{x_l + x_u}{2} \right|}{\left| \dfrac{x_l + x_u}{2} \right|} \prec 100\% \end{array} \right\}$$


12
16
14
16
15
14

Evaluation of Bisection Method

<u>Pros</u>

- Easy
- Always find root
- Number of iterations required to attain an absolute error can be computed a priori.

<u>Cons</u>

- Slow
- Know a and b that bound root
- Multiple roots
- No account is taken of $f(x_l)$ and $f(x_u)$, if $f(x_l)$ is closer to zero, it is likely that root is closer to x_l.

How Many Iterations will It Take?

- Length of the first Interval $L_o = b - a$
- After 1 iteration $L_1 = L_o/2$
- After 2 iterations $L_2 = L_o/4$
- …
- After k iterations $L_k = L_o/2^k$ or $k = ln_2(L_o/L_k)$

$$\varepsilon_a \le \frac{L_k}{x} \times 100\% \qquad \varepsilon_a \le \varepsilon_s$$

- If the absolute magnitude of the error is $\dfrac{\varepsilon_s \cdot x}{100\%} = 10^{-4}$

  and $L_o = 2$, how many iterations will you have to do to get the required accuracy in the solution?

$$10^{-4} = \frac{2}{2^k} \quad \Rightarrow 2^k = 2 \times 10^4 \quad \Rightarrow k \cong 14.3 = 15$$

Problem Solving

Problem Statement. Use bisection method to locate the root of $f(x) = x^{10} - 1$ between $x = 0$ and 1.3.

Solution. Using bisection, the results can be summarized as

$\varepsilon_a = |(x_l - x_r)/x_r|*100\%$ *and* $\varepsilon_t = |(Root - x_r)/Root|*100\%$

Iteration	x_l	x_u	$x_r=(x_l+x_u)/2$	ε_a (%)	ε_t (%)
1	0	1.3	0.65	100.0	35
2	0.65	1.3	0.975	33.3	2.5
3	0.975	1.3	1.1375	14.3	13.8
4	0.975	1.1375	1.05625	7.7	5.6
5	0.975	1.05625	1.015625	4.0	1.6

Thus, after five iterations, the true error is reduced to less than 2%.

Problem Solving

- Use bisection method to determine the value of $\sqrt{3}$ between $x = 1$ and 2.
- Solution: Let, x = $\sqrt{3}$ *or,* $x^2 = 3$, *or* $f(x) = x^2 - 3 = 0$ *Root = 1.7320508*
- $f(1) = -2$ *(-ve) and* $f(2) = 1$ *(+ve)*

$\varepsilon_a = |(x_l - x_r)/x_r|*100\%$ *and* $\varepsilon_t = |(Root - x_r)/Root|*100\%$

Iteration	x_l	x_u	$x_r=(x_l+x_u)/2$	ε_a (%)	ε_t (%)
1	1	2	1.5	33	13
2	1.5	2	1.75	14.28	?
3	1.5	1.75	?	?	?
4	??	??	?	?	?
5	??	??	??	??	??

Thus, after five iterations, the absolute error is reduced to less than ?%.